Track E: Data protection and governance

Generated from the same content files as the app. Each module lists its roadmap months. Solutions are in the appendix.

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Track E: Data protection and governance

The law and the frameworks that govern people data, for the engineer who builds the platform.

E1 UK GDPR and special category data in HR

Roadmap: EY Talent Data Strategy role; Current work. Personal and special category data, lawful bases for talent data, Article 9 conditions, appropriate policy documents and DPIA screening.

E1-L1 Personal data, special category data and the two-part test

Concept

Everything a talent platform holds is **personal data**: names, grades, salaries, review scores, course completions, skills. It relates to an identifiable person, so the UK GDPR applies. The Data Protection Act 2018 sits beside it and adds UK-specific conditions. These lessons follow the ICO's guidance as of November 2026. Your data protection officer and legal team make the final call.

Some personal data needs more protection. The UK GDPR names nine **special categories** in Article 9:

- racial or ethnic origin
- political opinions
- religious or philosophical beliefs
- trade union membership
- genetic data
- biometric data, when you use it to identify someone uniquely
- health
- sex life
- sexual orientation

Three details catch engineers out.

Biometric data is narrow. A staff photo on a profile is not special category data. The same photo used to unlock a door by face recognition is.

Inferences count. A model that infers health, ethnicity or sexual orientation from other fields processes special category data, even when no column is called that.

Protected characteristics are a different list. Age and sex are protected under the Equality Act 2010, but they are not special categories. Sexual orientation and religion are both.

To use any personal data you need a **lawful basis** under Article 6. To use special category data you need a lawful basis **and** an **Article 9 condition**. This is the **two-part test**. Lessons E1-L2 and E1-L3 cover each part.

Why it matters

An HR analytics team built an attrition model from absence records, overtime and the free-text reasons managers gave for sick leave. The model's strongest feature was a word count of "stress" and "anxiety" in those reasons. Nobody had flagged the column, because it was called absence_note. It held health data, so the processing needed an Article 9 condition and a data protection impact assessment that nobody had done. The model was withdrawn for three months while the team removed the field and wrote the assessment.

Worked example: weak code

```
# Attribute register, talent platform
date_of_birth: {sensitive: true}
gender: {sensitive: true}
absence_note: {sensitive: false}
badge_photo: {sensitive: true}
union_rep: {sensitive: false}
```

1. **Line 2.** One yes/no flag mixes up two lists. Age is a protected characteristic, not a special category.
2. **Line 4.** Free text about sick leave holds health data.
3. **Line 6.** Being a union representative reveals trade union membership, a special category.

Worked example: strong code

```
# Attribute register, talent platform
date_of_birth: {special_category: null, protected_characteristic: age}
gender: {special_category: null, protected_characteristic: sex}
absence_note: {special_category: health, protected_characteristic: null}
badge_photo: {special_category: null, note: "biometric only if used to
  identify"}
union_rep: {special_category: trade_union_membership,
  protected_characteristic: null}
```

1. **Line 2.** Two separate fields: the GDPR list and the Equality Act list.
2. **Line 4.** Classified by what the content reveals, not by the column name.

3. **Line 5.** Records the condition that would change the classification.

Classify each attribute by what it reveals. Keep the special categories and the protected characteristics as two separate fields, and treat inferences as data.

Recap card

- **The nine special categories** Racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic, biometric (to identify), health, sex life, sexual orientation.
- **Is an inferred special category special category data?** Yes. A model that infers health or ethnicity processes special category data.
- **Age and sex** Protected characteristics under the Equality Act 2010. Not special categories.
- **The two-part test** An Article 6 lawful basis for all personal data, plus an Article 9 condition for special category data.

Practice

Exercise 1. Multiple choice (level 1)

Which attribute is special category data under Article 9 of the UK GDPR?

- a) Date of birth
- b) Salary band
- c) Trade union membership
- d) Home address

Exercise 2. Multiple choice (level 2)

A model predicts which employees are likely to go on long-term sick leave, from absence patterns and overtime. No input column holds health data. Does the processing involve special category data?

- a) No, because no input column is health data
- b) Yes, because the model infers information about health
- c) Only if the predictions are shown to managers
- d) Only if the model uses more than 1,000 records

Why?

- r1) Inferring a special category from other data counts as processing that category.
- r2) Absence data is always health data.

Exercise 3. Fill the gap (level 2)

Complete the two-part test for using special category data.

```
Part 1: a lawful basis under Article _____(1)
Part 2: a condition under Article _____(2)
```

Exercise 4. Write the function (level 3)

Write `is_special_category(field: dict) -> bool` for an attribute register. `field` has a `"category"` key, such as `"health"` or `"age"`, and may have a `"use"` key. Return True for the nine special categories. Biometric data counts only when `field.get("use") == "identify"`. Keep the category names in a module constant.

The category names are: `"racial_or_ethnic_origin"`, `"political_opinions"`, `"religious_or_philosophical_beliefs"`, `"trade_union_membership"`, `"genetic"`, `"biometric"`, `"health"`, `"sex_life"` and `"sexual_orientation"`.

Your code must pass these checks:

- Return True for each of the nine special categories.
- Age and sex are protected characteristics, not special categories. Salary and address are ordinary personal data.
- Biometric data is special category data only when you use it to identify someone uniquely.

Shape of the answer. The reference solution defines these names. Write the bodies yourself.

```python
# Module constant.
SPECIAL_CATEGORIES = ...

# True when an attribute is special category data under Article 9.
def is_special_category(field: dict) -> bool: ...
```

Starter code:

```python
def is_special_category(field):
    return field["category"] in {"health", "age", "biometric"}
```

Exercise 5. Architecture design (level 4)

Design the classification fields for a talent attribute register that feeds access rules and DPIA screening.

Options: special_category and protected_characteristic as separate fields, One sensitive: true/false flag.

1. Fields for sensitivity
2. How an attribute is classified

3. Model outputs

Sources

- UK GDPR Article 9 lists nine special categories; biometric data is special category only when used to identify someone uniquely; inferring a special category means processing special category data; age is a protected characteristic but not special category data; special category data needs a lawful basis and an Article 9 condition. (search-only: ICO guidance pages 'What is special category data?', 'Biometric recognition: key data protection concepts' and 'What about fairness, bias and discrimination?' (ico.org.uk), seen through search results, November 2026. ico.org.uk was not reachable from the build environment.)
- The Data Protection Act 2018 supplements the UK GDPR with UK-specific conditions in Schedule 1. (search-only: ICO guidance 'What are the conditions for processing?' (ico.org.uk), seen through search results, November 2026.)

E1-L2 A lawful basis for each use of talent data

Concept

Article 6 of the UK GDPR gives **six lawful bases**. Each use of personal data needs one that fits it:

- **consent**
- **contract**, to perform a contract with the person, such as paying their salary
- **legal obligation**, such as tax reporting or right-to-work checks
- **vital interests**, to protect someone's life
- **public task**
- **legitimate interests**

Since 5 February 2026, the Data (Use and Access) Act 2025 adds **recognised legitimate interests**: a short, fixed list of purposes, such as safeguarding and crime prevention, that skip the balancing test. Commercial uses are not on it.

Employers rarely rely on consent. The ICO's view is that the power imbalance between employer and worker makes it hard for consent to be freely given. A worker who fears saying no has no genuine choice. Use another basis.

Workforce analytics usually rests on **legitimate interests**. The ICO describes a **three-part test**, recorded in a **legitimate interests assessment** (LIA) before processing starts:

1. **Purpose**: is there a legitimate interest?
2. **Necessity**: does the processing serve it, and is there a less intrusive way?
3. **Balancing**: do the person's interests, rights and freedoms override yours? Ask what a reasonable person would expect.

Two principles from Article 5 shape the platform. **Purpose limitation**: a basis covers the purposes you specified. Payroll under contract does not cover an attrition model. A new purpose needs its own check. **Data minimisation**: hold only the data each purpose needs.

Why it matters

A firm launched an engagement survey and stated that consent was its lawful basis. Two years later it fed the answers into a promotion-readiness model. Staff complained that they had never agreed to that use, and some withdrew consent, which removed their records from every analysis. A legitimate interests assessment for each purpose, done before launch, would have shown that the promotion use failed the balancing test.

Worked example: weak code

```
  # Processing register, talent platform
1 payroll: {lawful_basis: consent}
  engagement_survey: {lawful_basis: consent}
2 attrition_model: {lawful_basis: same as payroll}
```

1. **Line 2.** Workers cannot refuse payroll. Consent cannot be freely given here.
2. **Line 4.** A new purpose needs its own basis check. Payroll's basis does not stretch to it.

Worked example: strong code

```
  # Processing register, talent platform
1 payroll: {lawful_basis: contract}
  tax_reporting: {lawful_basis: legal_obligation}
2 engagement_survey: {lawful_basis: legitimate_interests, lia: LIA-007}
3 attrition_model: {lawful_basis: legitimate_interests, lia: LIA-012,
    fields: minimised}
```

1. **Line 2.** Paying salary performs the employment contract.
2. **Line 4.** Each use of legitimate interests points to its recorded assessment.
3. **Line 5.** Its own assessment, and only the fields the model needs.

Give each purpose its own lawful basis, avoid consent for workers, record a legitimate interests assessment for each use of legitimate interests, and keep only the fields each purpose needs.

Recap card

- **The six lawful bases** Consent, contract, legal obligation, vital interests, public task, legitimate interests.
- **Consent for employee data** Rarely appropriate. The power imbalance makes it hard to give freely. Use another basis.
- **Legitimate interests assessment** Purpose, necessity, balancing, recorded before processing starts.
- **Purpose limitation** A basis covers the purposes you specified. A new purpose, such as an attrition model, needs its own check.
- **Recognised legitimate interests** Added by the Data (Use and Access) Act 2025, in force 5 February 2026. A fixed list of purposes with no balancing test. Not for commercial uses.

Practice

Exercise 1. Multiple choice (level 1)

Which lawful basis fits paying an employee's salary?

- a) Consent
- b) Contract
- c) Vital interests
- d) Public task

Exercise 2. Multiple choice (level 2)

A team wants to use consent as the lawful basis for a voluntary skills survey of its own employees. What does the ICO's guidance say?

- a) Consent is the safest basis for any employee data
- b) Employers rarely rely on consent, because the power imbalance makes it hard to give freely
- c) Consent is banned for employees
- d) Consent is required for any survey

Why?

- r1) A worker who fears refusing has no genuine choice, so consent is hard to show as freely given.

- r2) Consent expires after one year.

Exercise 3. Put in order (level 2)

Put the steps of a legitimate interests assessment in order.

- Balancing: weigh the person's interests, rights and freedoms against yours
- Purpose: identify the legitimate interest
- Record the outcome before processing starts
- Necessity: show the processing serves it and nothing less intrusive would

Exercise 4. Fill the gap (level 3)

Complete the two principles that shape a talent platform.

```
Purpose _____(1): a lawful basis covers the purposes you specified. A new
    purpose needs its own check.
Data _____(2): hold only the data each purpose needs.
```

Exercise 5. Architecture design (level 4)

Choose the lawful basis for each use of a talent platform's data.

Options: Legal obligation, Consent, Legitimate interests.

1. Right-to-work checks before someone starts
2. Recording mandatory training completions
3. A new attrition prediction model

Sources

- Six Article 6 lawful bases; a new purpose needs a lawful basis that fits it; Article 5(1)(c) data minimisation and purpose limitation. (search-only: ICO guidance 'Principle (b): Purpose limitation', 'Principle (c): Data minimisation' and 'A guide to lawful basis' (ico.org.uk), seen through search results, November 2026.)
- Consent is not usually appropriate for employers because of the imbalance of power; not banned where refusing has no adverse consequences. (search-only: ICO guidance 'Data protection and monitoring workers' and 'Data protection and workers' health information' (ico.org.uk), seen through search results, November 2026.)
- Legitimate interests three-part test: purpose, necessity, balancing; reasonable expectations judged objectively; record in an LIA before processing. (search-only: ICO guidance 'What is the legitimate interests

basis?' and 'Legitimate interests' (ico.org.uk), seen through search results, November 2026.)

- Recognised legitimate interests came into force on 5 February 2026 (Data (Use and Access) Act 2025, Commencement No. 6, SI 2026/82); listed purposes skip the balancing test; not for commercial processing. (search-only: Commentary from Alston & Bird (January 2026), Pearl Cohen and bratby.law, seen through search results, November 2026. legislation.gov.uk was not reachable from the build environment.)

E1-L3 Article 9 conditions, policy documents and DPIA screening in HR

Concept

Article 9(2) of the UK GDPR lists ten conditions for special category data. Two carry most HR work, and in the UK both need an extra condition from **Schedule 1 of the Data Protection Act 2018**.

Article 9(2)(b), employment, social security and social protection. It covers duties and rights in employment law, such as recording sickness absence or making reasonable adjustments. The UK condition is Schedule 1, Part 1, **paragraph 1**. It does not cover purely contractual rights.

Article 9(2)(g), substantial public interest. It needs one of the 23 conditions in Schedule 1, **paragraphs 6 to 28**. Diversity monitoring usually relies on **paragraph 8, equality of opportunity or treatment**. Paragraph 8 supports reviewing equality across groups. It cannot support a decision about one person, so keep monitoring data apart from promotion or hiring decisions.

Both routes need an **appropriate policy document** (APD). The ICO says it needs no set form. It names the Schedule 1 condition, explains how you meet the principles, and sets your retention and erasure rules. One APD can cover several purposes. Your record of processing activities also names the condition.

Before you start processing likely to result in **high risk**, carry out a **data protection impact assessment** (DPIA) under Article 35. Three cases always need one, including **large-scale special category data** and extensive profiling with significant effects. The ICO's list adds others, such as innovative technology like AI combined with another risk factor. Two risk

factors usually mean a DPIA. If you cannot reduce a high risk, consult the ICO before you start.

Why it matters

A retailer used its equal opportunities survey, collected under paragraph 8, to pick candidates for a leadership programme that favoured under-represented groups. The survey data now drove decisions about individuals, which paragraph 8 does not support. The programme paused while the team rebuilt it on aggregate statistics and a separate, voluntary application form.

Worked example: weak code

```
  1 # Special category register
❶ sickness_absence: {article_9: 9(2)(b)}
❷ ethnicity: {article_9: 9(2)(g), used_for: [diversity_report,
    promotion_shortlist]}
❸ dpia: not needed, internal data only
```

1. **Line 2.** No Schedule 1 condition and no appropriate policy document.
2. **Line 3.** Paragraph 8 supports monitoring, not decisions about individuals.
3. **Line 4.** Large-scale special category data needs a DPIA, internal or not.

Worked example: strong code

```
  1 # Special category register
❶ sickness_absence: {article_9: 9(2)(b), schedule_1: "Part 1 para 1", apd:
    APD-HR-01}
❷ ethnicity: {article_9: 9(2)(g), schedule_1: "Part 2 para 8", apd: APD-
    HR-01, used_for: [diversity_report]}
❸ dpia: DPIA-014, screened for large-scale special category data and
    profiling
```

1. **Line 2.** Article 9 condition, Schedule 1 condition and the policy document, side by side.
2. **Line 3.** Monitoring only. No individual decisions.
3. **Line 4.** The screening reason is recorded with the assessment.

Pair each Article 9 condition with its Schedule 1 condition and an appropriate policy document, keep monitoring data away from individual decisions, and screen every new use for a DPIA.

Recap card

- **Employment condition in the UK** Article 9(2)(b) plus Schedule 1, Part 1, paragraph 1 of the DPA 2018, with an appropriate policy document.

- **Diversity monitoring** Article 9(2)(g) plus Schedule 1 paragraph 8, equality of opportunity or treatment. Never for decisions about one person.
- **Appropriate policy document** Names the Schedule 1 condition, how you meet the principles, and retention and erasure. One APD can cover several purposes.
- **When a DPIA is mandatory** Before high-risk processing, always for large-scale special category data and extensive profiling with significant effects. Consult the ICO if high risk remains.

Practice

Exercise 1. Multiple choice (level 1)

Which conditions does diversity monitoring of employees usually rely on?

- a) Article 9(2)(g) with Schedule 1 paragraph 8, equality of opportunity or treatment
- b) Article 9(2)(b) with Schedule 1 paragraph 1
- c) Consent from each employee
- d) No condition, because the data is aggregated afterwards

Exercise 2. Multiple choice (level 2)

A team wants to use the ethnicity answers from the equal opportunities survey to shortlist candidates for a promotion programme. What is the problem?

- a) None, the data is already collected
- b) Paragraph 8 supports monitoring equality across groups, not decisions about one person
- c) Ethnicity is not special category data
- d) The survey must be repeated each year

Why?

- r1) The ICO says the equality condition cannot be relied on when processing is for decisions about a particular individual.
- r2) Survey data expires after six months.

Exercise 3. Fill the gap (level 2)

Complete the documentation rule for the employment condition.

```
Article 9(2)(b) in the UK needs Schedule 1, Part 1, paragraph _____(1)
```

```
2 and an appropriate _____(2) document that names the condition and the
    retention rules.
```

Exercise 4. Write the function (level 3)

Write `check_register(entries: list[dict]) -> list[str]` for a processing register. Each entry has `"attribute"`, `"special"` (a bool), `"lawful_basis"`, `"condition"`, `"schedule1"` and `"apd"` (a bool). `"condition"` and `"schedule1"` may be None. Return one message per problem, in register order:

- `"<attribute>: no lawful basis"` when `lawful_basis` is not one of the six bases (`"consent"`, `"contract"`, `"legal_obligation"`, `"vital_interests"`, `"public_task"`, `"legitimate_interests"`)
- `"<attribute>: no Article 9 condition"` when the attribute is special and `condition` is None
- `"<attribute>: no Schedule 1 condition"` when `condition` is `"9(2)(b)"` or `"9(2)(g)"` and `schedule1` is None
- `"<attribute>: no appropriate policy document"` when `condition` is one of those two and `apd` is False

Keep the bases and the two conditions in module constants.

What you need to know.

`entry["schedule1"] is None` tests for a missing value without treating an empty string as missing. A list collects messages in order:

```
1 notes = []
2 notes.append("first")
3 notes.append("second")   # ["first", "second"]
```

Your code must pass these checks:

- A complete special entry and an ordinary entry raise no problems.
- Flag a lawful basis that is not one of the six.
- A special attribute needs an Article 9 condition.
- 9(2)(b) and 9(2)(g) need a Schedule 1 condition and an appropriate policy document.
- Return the problems in register order, not sorted.

Shape of the answer. The reference solution defines these names. Write the bodies yourself.

```
1 # Module constant.
2 LAWFUL_BASES = ...
3
4 # Module constant.
5 NEEDS_SCHEDULE_1 = ...
6
```

```python
# List the documentation gaps in a processing register.
def check_register(entries: list[dict]) -> list[str]: ...
```

Starter code:

```python
def check_register(entries):
    return []
```

Exercise 5. Architecture design (level 4)

Screen a new use of talent data for a DPIA: a model that ranks all 40,000 employees for promotion readiness, using performance ratings, absence history and inferred health risk.

Options: Yes: large-scale special category data and profiling with significant effects, No: the data is internal and already held.

1. Is a DPIA needed?
2. When
3. If a high risk remains after mitigation

Sources

- Article 9(2)(b) relies on DPA 2018 Schedule 1 Part 1 paragraph 1 and needs an appropriate policy document; the employment condition does not cover purely contractual rights; APD needs no set form, outlines condition, principles procedures and retention; one APD can cover several purposes; record of processing activities must note the condition. (search-only: ICO guidance 'What are the conditions for processing?', 'What are the rules on special category data?' and 'Data protection and monitoring workers' (ico.org.uk), seen through search results, November 2026.)
- 23 substantial public interest conditions in Schedule 1 paragraphs 6 to 28; paragraph 8 equality of opportunity or treatment supports monitoring with Article 9(2)(g); it cannot be relied on for decisions about a particular individual; APD required. (search-only: ICO guidance 'What are the substantial public interest conditions?' and 'What about fairness, bias and discrimination?' (ico.org.uk), seen through search results, November 2026. The Data (Use and Access) Act 2025 added a substantial public interest condition, not checked here.)
- Article 35(3): large-scale special category data, systematic extensive profiling with significant effects and large-scale public monitoring always need a DPIA; ICO list includes innovative technology such as AI when combined with another criterion; two screening criteria usually indicate a DPIA; consult the ICO if high risk cannot be mitigated. (search-only: ICO

guidance 'When do we need to do a DPIA?' and 'Examples of processing likely to result in high risk' (ico.org.uk), seen through search results, November 2026.)

E2 DAMA-DMBOK applied to talent data

Roadmap: EY Talent Data Strategy role; Current work. The eleven knowledge areas, governance at the hub, and an evidence-based maturity baseline across talent domains.

E2-L1 The DAMA wheel, eleven knowledge areas around governance

Concept

The **DAMA-DMBOK** (Data Management Body of Knowledge, second edition, revised in 2024) gives data teams a shared map. It splits data management into **eleven knowledge areas**, drawn as a wheel.

Data governance sits at the hub. It decides and oversees how the other ten are run. DMBOK2 defines it as the exercise of authority and control, planning, monitoring and enforcement, over the management of data assets.

The ten spokes, with a talent example for each:

Knowledge area	Talent example
Data architecture	Bronze, silver and gold layers for HR data
Data modelling and design	The person, position and organisation entities
Data storage and operations	Lakehouse tables, backups, retention jobs
Data security	Masking salary and health fields
Data integration and interoperability	Feeds from the HR system and the learning platform
Document and content management	Contracts, review forms, scanned certificates
Reference and master data	One golden record per person, job codes, grades
Data warehousing and business intelligence	Headcount and attrition reports
Metadata management	Glossary entries, lineage, owners
Data quality	Rules for missing managers or duplicate employee IDs

Two pairs get confused. **Reference data** is a controlled list of codes, such as grades or country codes. **Master data** is the agreed record of a core entity, such as the person. **Metadata** describes data: definitions, owners, lineage. **Master data** is the data itself.

The wheel is a checklist, not an org chart. A talent platform touches all eleven areas. When something breaks, name the area first, then the owner. A duplicate employee is a master data problem, not a reporting bug.

Why it matters

A firm's headcount report disagreed with payroll by 312 people. The BI team spent six weeks tuning report filters. The cause was in master data: contractors re-hired through an agency got a second employee ID, and nothing matched them to their first record. Naming the knowledge area on day one would have sent the problem to the master data owner.

Worked example: weak code

```
# Issue log, talent platform
- issue: headcount differs from payroll by 312
  owner: BI team
  fix: adjust the report filter
```

1. **Line 3.** The report shows the symptom. The cause sits in another knowledge area.
2. **Line 4.** Fixing the symptom hides the duplicate records from every other consumer.

Worked example: strong code

```
# Issue log, talent platform
- issue: headcount differs from payroll by 312
  knowledge_area: reference and master data
  root_cause: re-hired contractors get a second employee ID
  owner: people master data owner
  fix: match rule on national insurance number and date of birth, then
   merge
```

1. **Line 3.** Name the area first. It points at the right owner.
2. **Line 5.** Governance names an owner for each area and domain.

Name the knowledge area before you name the fix. Governance at the hub assigns an owner to each area, so problems reach the person who can solve them.

Recap card

- **DAMA wheel** Data governance at the hub, ten knowledge areas around it. DMBOK2, revised in 2024.
- **DMBOK2 definition of data governance** The exercise of authority and control (planning, monitoring and enforcement) over the management of data assets.
- **Reference data, master data, metadata** Reference: lists of codes. Master: the agreed record of a core entity. Metadata: data about data.
- **First move on any data problem** Name the knowledge area, then its owner, then the fix.

Practice

Exercise 1. Multiple choice (level 1)

Which knowledge area sits at the hub of the DAMA wheel?

- a) Data architecture
- b) Data governance
- c) Data quality
- d) Metadata management

Exercise 2. Multiple choice (level 2)

Re-hired contractors get a second employee ID, so headcount is overstated. Which knowledge area owns the root cause?

- a) Data warehousing and business intelligence
- b) Reference and master data
- c) Document and content management
- d) Data storage and operations

Why?

- r1) Master data keeps one agreed record per core entity, such as a person.
- r2) Reference data controls the format of employee IDs.

Exercise 3. Fill the gap (level 2)

Complete the two definitions that are easy to confuse.

```
_____(1) data: a controlled list of codes, such as grades or country codes
_____(2): data about data, such as definitions, owners and lineage
```

Exercise 4. Architecture design (level 3)

Send each talent data problem to its knowledge area.

Options: Data security, Data quality, Data warehousing and business intelligence.

1. Analysts can read salaries they should not see
2. Two teams define 'active employee' differently
3. 12% of records have no manager

Exercise 5. Architecture design (level 4)

You start a new governance role on a talent platform. Choose how to use the DAMA wheel in your first month.

Options: Map all eleven areas to the six talent domains, then pick where to start, Start with the area you know best and add others later.

1. Scope
2. Owners
3. How the wheel is used

Sources

- DAMA-DMBOK2 has eleven knowledge areas with data governance at the centre of the wheel: data architecture, data modelling and design, data storage and operations, data security, data integration and interoperability, document and content management, reference and master data, data warehousing and business intelligence, metadata management and data quality. A 2024 revision kept the framework and areas. (search-only: Secondary summaries (Dawiso glossary, OvalEdge, a reproduced DMBOK2 table of contents), seen through search results, November 2026. www.dama.org was not reachable from the build environment, and the book was not checked.)
- DMBOK2 section 3.1 (p. 67) defines data governance as the exercise of authority and control (planning, monitoring, and enforcement) over the management of data assets. The older DAMA Dictionary adds 'shared decision making'. (search-only: Government of Canada GCcollab wiki 'Data Governance' citing DMBOK2, and Perficient (2012) citing the DAMA Dictionary, seen through search results, November 2026.)

E2-L2 A maturity baseline you can defend

Concept

A **maturity baseline** records how well each knowledge area is run today, so you can set targets and show progress. DMBOK2 describes a data management maturity assessment. Sources differ on the exact level names, so agree one scale with your stakeholders and write it down. A common version runs from 0 to 5:

- **0** no capability
- **1** initial: ad hoc, depends on individuals
- **2** repeatable: some process, applied unevenly
- **3** defined: a documented standard, followed across teams
- **4** managed: measured, with targets and reviews
- **5** optimised: improved continuously from the measures

Each level builds on the last, so none can be skipped.

Score a **grid**, not a single number: knowledge areas down the side, talent domains across the top. A talent platform has six domains: workforce and organisation, recruitment, learning, performance, mobility and skills. Data quality can be at 3 for workforce data and at 1 for skills.

Three rules make a baseline you can defend.

1. **Evidence for every score.** A score without a document, a metric or a log is an opinion. "Level 3: quality rules documented in DQ-STD-02 and run nightly" survives a challenge.
2. **Report the weakest area, not the mean.** An average of 3 can hide a 0 in data security. One weak area can sink a domain.
3. **Integers only.** A level is a stage you have reached. A score of 2.5 says you have not agreed which stage you are at.

The baseline is a snapshot. Rescore at fixed intervals with the same scale, so changes mean progress, not a new yardstick.

Why it matters

A firm reported its talent data maturity as 3.1 out of 5 to its board and won budget for an AI pilot. The pilot stalled in week two: nobody owned access to performance data, which scored 0 for data security in the performance domain. The average had hidden it. A grid reporting the weakest area per domain would have put security ahead of the pilot.

Worked example: weak code

```
# Maturity baseline, talent data
overall: 3.1
data_security: 2.5
evidence: workshop with the HR leadership team
```

1. **Line 2.** One mean for the whole platform hides the weakest area.
2. **Line 3.** A fractional level means the stage was never agreed.
3. **Line 4.** Opinions, not artefacts. Nothing an auditor can check.

Worked example: strong code

```
# Maturity baseline, talent data, scale DMMA-0-5 v1
performance:
  data_security: {score: 0, evidence: "no access model; analysts read raw reviews"}
  data_quality: {score: 2, evidence: "rules in DQ-PERF-01, run by hand monthly"}
  weakest: {area: data_security, score: 0}
workforce:
  data_security: {score: 3, evidence: "role model SEC-STD-01, quarterly access review log"}
  weakest: {area: metadata, score: 1}
```

1. **Line 1.** The scale is named and versioned, so rescoring uses the same yardstick.
2. **Line 3.** Every score cites evidence.
3. **Line 5.** Each domain reports its weakest area, not its average.

Score a grid of knowledge areas by talent domain, in whole levels, with evidence for each score, and report the weakest area per domain.

Recap card

- **A common 0 to 5 maturity scale** 0 none, 1 initial, 2 repeatable, 3 defined, 4 managed, 5 optimised. Agree and version the names with stakeholders.
- **The baseline grid** Knowledge areas by talent domain: workforce, recruitment, learning, performance, mobility, skills.
- **Summarise a domain by** Its weakest area and score. A mean hides gaps.
- **A defensible score** A whole level with cited evidence: a document, a metric or a log.

Practice

Exercise 1. Multiple choice (level 1)

Data quality rules for the learning domain are documented in a standard and followed by every team, but nobody measures them yet. Which level fits, on the 0 to 5 scale in this lesson?

- a) 1 initial
- b) 3 defined
- c) 4 managed
- d) 5 optimised

Exercise 2. Multiple choice (level 2)

The performance domain scores 4, 4, 3, 4 and 0 across five knowledge areas. What should the baseline report for the domain?

- a) 3, the mean
- b) 4, the most common score
- c) 0, with the area that scored it
- d) Nothing until every area reaches 3

Why?

- r1) One weak area can sink a domain, and a mean hides it.
- r2) The lowest score is always data security.

Exercise 3. Put in order (level 2)

Put the steps of building a maturity baseline in order.

- Agree and version the scale with stakeholders
- Score each cell in whole levels, citing evidence
- Lay out the grid: knowledge areas by talent domain
- Report the weakest area per domain
- Rescore at a fixed interval on the same scale

Exercise 4. Write the function (level 3)

Write `weakest_by_domain(scores: dict[str, dict[str, int]]) -> dict[str, tuple[str, int]]`. `scores` maps each talent domain to a dict of knowledge area to score. Return, for each domain, the weakest area and its score. When two areas tie, pick the one that comes first alphabetically.

Raise `ValueError` if any score is not a whole number from 0 to 5, or if a domain has no scores. A `bool` does not count as a whole number.

What you need to know.

`isinstance(value, int)` checks a type. In Python, `True` is also an `int`, so check for `bool` first:

```
isinstance(3, int)       # True
isinstance(2.5, int)     # False
isinstance(True, bool)   # True, and isinstance(True, int) is True too
```

`min` takes a `key` function. A tuple key compares item by item: `min([("b", 1), ("a", 1)], key=lambda p: (p[1], p[0]))` returns `("a", 1)`.

Your code must pass these checks:

- Return the lowest-scoring area and its score for each domain.
- When areas tie, pick the one that comes first alphabetically.
- Raise ValueError for a score outside 0 to 5, a fraction, a bool, or a domain with no scores.

Shape of the answer. The reference solution defines these names. Write the bodies yourself.

```
# Module constant.
MIN_LEVEL = ...

# Module constant.
MAX_LEVEL = ...

# The weakest knowledge area and its score for each talent domain.
def weakest_by_domain(
    scores: dict[str, dict[str, int]],
) -> dict[str, tuple[str, int]]: ...
```

Starter code:

```
def weakest_by_domain(scores):
    return {domain: min(areas.items()) for domain, areas in
    scores.items()}
```

Exercise 5. Architecture design (level 4)

Plan how you present the first talent data maturity baseline to HR leadership.

Options: A grid of areas by domain, with the weakest area per domain highlighted, One overall maturity score out of 5.

1. Headline
2. Evidence
3. What the baseline leads to

Sources

- DMBOK2 describes a data management maturity assessment; a common scale runs 0 no capability, 1 initial or ad hoc, 2 repeatable, 3 defined, 4 managed, 5 optimised; levels cannot be skipped. (unverified: Practitioner and consultancy sources (datacrossroads.nl, DAMA chapter presentations on SlideShare), seen through search results, November 2026. Sources disagree on names and numbering, and DMBOK2 itself was not checked.)
- The six talent domains (workforce and organisation, recruitment, learning, performance, mobility, skills) come from the role brief, not from DAMA. (verified: content/roles/ey-talent.yaml, the job description in the brief.)

E2-L3 Governance at the hub, decision rights, policies and escalation

Concept

Governance and management are different jobs. **Data governance** decides and oversees: who may decide what, which rules apply, and whether they are followed. **Data management** does the work: builds the pipelines, runs the quality checks, grants the access. DMBOK2 frames governance as setting decision rights over data. A platform team that both writes the rules and marks its own work has no oversight.

Start with **decision rights**. For each kind of decision, name who decides:

- what a term means, such as "active employee" or "regretted leaver"
- who may see a field, such as salary or a performance rating
- what quality threshold a dataset must meet before it is published
- whether a new use of data goes ahead

Then write the rules at three levels:

- A **policy** says what must be true, and why. "Access to performance data follows the principle of least privilege."
- A **standard** makes it measurable. "Ratings are masked for every role except HR business partners and the employee's own line chain."
- A **procedure** says how. "Request access through the form. The data owner approves within five working days."

Policies change rarely and need senior sign-off. Standards change when the platform does. Procedures change often.

Finally, an **escalation path**. Most issues end with a data steward. Those that cross domains, or set precedent, go to the data owner, then to a governance council. Each decision is written down in a decision log (lesson B18-L3), so the next person finds the reason, not only the rule.

GOV-02 in your EY role plan turns these rights into a RACI.

Why it matters

Two directors argued for a quarter over whether contractors counted in "headcount". The data team changed the dashboard each time one of them complained. Nobody had the right to decide, so the definition flipped four times and trust in every people report fell. A named decision owner and one log entry settled it in a week.

Worked example: weak code

```
1 # Governance, talent platform
2 rules: the data platform team decides and enforces everything
3 access_rule: be careful with sensitive data
4 disputes: raise in the weekly team meeting
```

1. **Line 2.** The same team writes the rules and marks its own work. No oversight.
2. **Line 3.** Not measurable. Nobody can test it.
3. **Line 4.** No owner, no record, so the same dispute comes back.

Worked example: strong code

```
1 # Governance, talent platform
2 decision_rights:
3   definitions: {decides: data owner per domain, advises: data steward}
4   access_to_performance_data: {decides: performance data owner}
5 policy: POL-07 least privilege for performance data
6 standard: STD-07 ratings masked except HR business partners and line chain
7 procedure: PRC-07 request form, owner approves within 5 working days
8 escalation: [data steward, data owner, governance council]
9 decisions: logged in DEC-LOG with the reason and the date
```

1. **Line 3.** One named decider per kind of decision.
2. **Line 6.** The standard makes the policy testable.
3. **Line 8.** A fixed path, so disputes end with a decision.

Separate deciding from doing. Name a decider for each kind of decision, write rules as policy, standard and procedure, and log each decision with its reason.

Recap card

- **Governance against management** Governance decides and oversees. Management does the work.
- **Decision rights to name first** Definitions, access to sensitive fields, quality thresholds, and new uses of data.
- **Policy, standard, procedure** Policy: what and why. Standard: measurable. Procedure: how. Each changes at its own pace.
- **Escalation path** Data steward, data owner, governance council. Log every decision with its reason.

Practice

Exercise 1. Multiple choice (level 1)

Which of these is a data governance activity rather than a data management one?

- a) Running the nightly quality checks
- b) Deciding who may approve access to salary data
- c) Building the silver layer for learning records
- d) Granting a role in a workspace

Exercise 2. Multiple choice (level 2)

Which statement is a standard, not a policy or a procedure?

- a) Access to performance data follows least privilege.
- b) Ratings are masked for every role except HR business partners and the line chain.
- c) Request access through the form, and the owner approves within five working days.
- d) Be careful with sensitive data.

Why?

- r1) A standard makes a policy measurable, so you can test whether it holds.
- r2) A standard says who signs it off.

Exercise 3. Fill the gap (level 2)

Complete the three levels of rules.

```
_____(1): what must be true, and why. Changes rarely, senior sign-off.
_____(2): makes it measurable. Changes when the platform does.
```

```
3 _____(3): how to do it. Changes often.
```

Exercise 4. Put in order (level 3)

A dispute over what "regretted leaver" means affects two domains. Put the escalation steps in order.

- The data owners of both domains try to agree
- Update the glossary entry and tell the report owners
- Record the decision and its reason in the decision log
- The data steward gathers the definitions in use and their impact
- Without agreement, the governance council decides

Exercise 5. Architecture design (level 4)

Set up governance for the talent platform's first domain, performance data.

Options: The performance data owner, a named HR leader, advised by a steward, The data platform team, because it builds the tables.

1. Who decides what a performance rating means
2. The access rule
3. Disputes

Sources

- DMBOK2 frames data governance as authority and control over data management, establishing decision rights over data. (search-only: GCcollab wiki 'Data Governance' citing DMBOK2 section 3.1, and summaries of DMBOK2 chapter 3, seen through search results, November 2026.)
- Policy, standard and procedure as three levels of rules, and the steward, owner, council escalation path, are common governance practice, not quotations from DMBOK2. (unverified: General practice. Not attributed to a specific DMBOK2 passage.)

Track E best-practice checklist

☐ Treat every talent attribute as personal data. Flag special category data, including what a model infers.

☐ Every use of personal data has a lawful basis. Every use of special category data also has an Article 9 condition.

☐ Employers rarely rely on consent. Record a legitimate interests assessment when you use legitimate interests.

☐ Keep an appropriate policy document for the employment and substantial public interest conditions.

☐ Screen every new use of people data for a DPIA. Large-scale special category data always needs one.

☐ Govern with a frame everyone shares. Map each knowledge area to the talent domains, with an owner.

☐ Score maturity with evidence, per knowledge area and domain. Never average a gap away.

☐ The legal team and the data protection officer decide on compliance. These lessons teach you to ask the right questions.

Appendix: solutions and answer keys

Attempt each exercise before you read its solution. Track D answer keys are here too.

E1-L1 Personal data, special category data and the two-part test

E1-L1 exercise 1

Hints. (1) There are nine special categories. (2) Think of what membership of an organisation can reveal about someone. (3) Trade union membership is on the list.

Answer. c) Trade union membership

Trade union membership is one of the nine special categories. The others are racial or ethnic origin, political opinions, religious or philosophical beliefs, genetic data, biometric data used to identify someone, health, sex life and sexual orientation.

Common wrong answers.

- `a`: Age is a protected characteristic under the Equality Act 2010, but it is not one of the nine special categories.
- `b`: Salary is personal data, and confidential, but it is not a special category.
- `d`: An address is personal data, not special category data.

E1-L1 exercise 2

Hints. (1) What does the model output say about each person? (2) Does inferring a category count as processing it? (3) The ICO treats inferred special category data as special category data.

Answer. b) Yes, because the model infers information about health Because: r1) Inferring a special category from other data counts as processing that category.

The ICO's guidance says that inferring health, ethnicity, beliefs or sexual orientation means you are processing special category data and need an Article 9 condition.

Common wrong answers.

- `a`: Inferences count. A prediction about health is health data.

- `c`: Producing the inference is processing, whether or not anyone sees it.
- `d`: Scale matters for whether you need a DPIA, not for whether data is special category.
- `b|r2`: Right answer, wrong reason. A plain count of days absent is not health data. The model's inference about illness is.

E1-L1 exercise 3

Hints. (1) One article applies to all personal data. (2) The other applies only to special categories. (3) Article 6, then Article 9.

Answer.

```
Part 1: a lawful basis under Article 6
Part 2: a condition under Article 9
```

Every use of personal data needs an Article 6 lawful basis. Special category data needs an Article 9 condition as well.

Common wrong answers.

- g1 = `9`; g2 = `6`: The other way round. Every use of personal data needs an Article 6 basis. Special category data also needs an Article 9 condition.

E1-L1 exercise 4

Hints. (1) Put the nine names in a frozenset at module level. (2) Handle biometric before the general check. (3) `field.get("use") == "identify"` is True only when the use is set to identify.

Answer.

```python
SPECIAL_CATEGORIES = frozenset(
    {
        "racial_or_ethnic_origin",
        "political_opinions",
        "religious_or_philosophical_beliefs",
        "trade_union_membership",
        "genetic",
        "biometric",
        "health",
        "sex_life",
        "sexual_orientation",
    }
)


def is_special_category(field: dict) -> bool:
    """True when an attribute is special category data under Article 9.

    Args:
        field: Register entry with a "category" and an optional "use".
```

```
21
22      Returns:
23          Whether the attribute needs an Article 9 condition.
24      """
25      category = field["category"]
26      if category == "biometric":
27          return field.get("use") == "identify"
28      return category in SPECIAL_CATEGORIES
```

A constant holds the list, and the one conditional category gets its own branch. Age and sex stay out: they belong to the Equality Act list.

E1-L1 exercise 5

Hints. (1) Do the two lists of sensitive attributes match? (2) Can a column name hide what a field holds? (3) Is an inference data?

Answer.

- Two fields, because the GDPR list and the Equality Act list overlap only in part.
- Classify by content: a free-text field can hold health data under any name.
- Inferred attributes are special category data, so they belong in the register with their own access rules.

Separate the two lists, classify by content, and register inferences, so access rules and DPIA screening start from the truth.

Common wrong answers.

- fields = `one`: A single flag cannot say whether an attribute needs an Article 9 condition or only a fairness check.
- basis = `name`: A column called absence_note held health data in this lesson's example.
- inferred = `skip`: Inferring a special category is processing it.

E1-L2 A lawful basis for each use of talent data

E1-L2 exercise 1

Hints. (1) What agreement does a salary come from? (2) The employment contract. (3) Contract.

Answer. b) Contract

Paying salary is necessary to perform the employment contract, so the contract basis fits.

Common wrong answers.

- `a`: The employee cannot refuse payroll, so consent would not be freely given. Payroll performs the contract.
- `c`: Vital interests protects someone's life. Salary payments do not.
- `d`: Public task covers official functions set out in law, not paying staff.

E1-L2 exercise 2

Hints. (1) Who holds the power in employment? (2) Can consent be freely given when refusing feels risky? (3) Look for another basis, such as legitimate interests.

Answer. b) Employers rarely rely on consent, because the power imbalance makes it hard to give freely Because: r1) A worker who fears refusing has no genuine choice, so consent is hard to show as freely given.

The ICO treats consent as rarely appropriate for employers, because a worker's choice is not genuinely free. Legitimate interests, with a recorded assessment, usually fits voluntary analytics.

Common wrong answers.

- `a`: Consent is the hardest basis to rely on for employees, and it can be withdrawn at any time.
- `c`: It is not banned. The ICO says it can work where refusing truly has no consequences, but that is rare.
- `d`: No lawful basis is required for a type of activity. You choose the one that fits the purpose.
- `b|r2`: Right answer, wrong reason. The problem is the imbalance of power, not an expiry date.

E1-L2 exercise 3

Hints. (1) What do you need before you can test necessity? (2) The balancing test weighs the interest you named against the person's. (3) The record comes last, before processing begins.

Answer.

1. Purpose: identify the legitimate interest
2. Necessity: show the processing serves it and nothing less intrusive would
3. Balancing: weigh the person's interests, rights and freedoms against yours
4. Record the outcome before processing starts

Purpose, necessity, balancing, then a written record before processing starts.

Common wrong answers.

- You cannot weigh an interest you have not named yet. Start with the purpose.
- Record the outcome once all three parts are done, before processing starts.

E1-L2 exercise 4

Hints. (1) Both principles come from Article 5. (2) One limits purposes, the other keeps data to a minimum. (3) `limitation`, then `minimisation`.

Answer.

```
1 Purpose limitation: a lawful basis covers the purposes you specified. A
    new purpose needs its own check.
2 Data minimisation: hold only the data each purpose needs.
```

Purpose limitation and data minimisation are Article 5 principles. Together they stop one dataset quietly serving every new idea.

Common wrong answers.

- g1 = `minimisation`; g2 = `limitation`: The other way round: purpose limitation, data minimisation.

E1-L2 exercise 5

Hints. (1) Which uses does the law require? (2) Which use is a new purpose? (3) A new purpose needs its own basis check.

Answer.

- Right-to-work checks are required by law, so legal obligation fits.
- Training records support the employment relationship. Record which basis you use and why.
- The attrition model is a new purpose. It needs its own basis check and assessment, and only the fields it uses.

Legal duties use legal obligation. Running the employment relationship uses contract or legitimate interests. A new analytics purpose gets its own assessment.

Common wrong answers.

- right_to_work = `consent`: The check is a legal duty. Consent would be misleading, because the worker cannot refuse.
- attrition = `inherit`: Purpose limitation: payroll's basis does not cover a new analytics purpose.

- attrition = `consent`: Consent rarely works for employees, and withdrawals would leave holes in the model.

E1-L3 Article 9 conditions, policy documents and DPIA screening in HR

E1-L3 exercise 1

Hints. (1) Monitoring equality is a substantial public interest. (2) Which Schedule 1 paragraph names equality? (3) Paragraph 8.

Answer. a) Article 9(2)(g) with Schedule 1 paragraph 8, equality of opportunity or treatment

Article 9(2)(g) with Schedule 1 paragraph 8 covers reviewing equality of opportunity or treatment between groups.

Common wrong answers.

- `b`: Paragraph 1 covers obligations and rights in employment law. Diversity monitoring uses the equality condition in paragraph 8.
- `c`: Consent is rarely appropriate for employees, and monitoring has its own condition.
- `d`: You process individual records before you aggregate them, so you need a condition.

E1-L3 exercise 2

Hints. (1) What does paragraph 8 allow you to do? (2) Is a shortlist a decision about a person? (3) Monitoring and decisions need separate data and separate conditions.

Answer. b) Paragraph 8 supports monitoring equality across groups, not decisions about one person Because: r1) The ICO says the equality condition cannot be relied on when processing is for decisions about a particular individual.

Keep equality monitoring data apart from decisions about individuals. Paragraph 8 does not cover them.

Common wrong answers.

- `a`: Purpose limitation: the data was collected for monitoring, under a condition that does not cover decisions.
- `c`: Racial or ethnic origin is a special category.
- `d`: Frequency is not the issue. The use is.

- `b|r2`: Right answer, wrong reason. Retention periods are set in your policy, but the problem here is using monitoring data for individual decisions.

E1-L3 exercise 3

Hints. (1) Part 1 of Schedule 1 starts with the employment condition. (2) APD stands for appropriate ... document. (3) Paragraph `1`, and `policy`.

Answer.

```
Article 9(2)(b) in the UK needs Schedule 1, Part 1, paragraph 1
and an appropriate policy document that names the condition and the
    retention rules.
```

The employment condition is Schedule 1, Part 1, paragraph 1, and it needs an appropriate policy document.

Common wrong answers.

- g1 = `8`: Paragraph 8 is equality of opportunity or treatment, a substantial public interest condition. The employment condition is paragraph 1.

E1-L3 exercise 4

Hints. (1) Loop over the entries once, appending messages as you go. (2) Check the basis, then the Article 9 condition, then the two documentation rules. (3) Only 9(2)(b) and 9(2)(g) need Schedule 1 and the policy document.

Answer.

```python
LAWFUL_BASES = frozenset(
    {
        "consent",
        "contract",
        "legal_obligation",
        "vital_interests",
        "public_task",
        "legitimate_interests",
    }
)
NEEDS_SCHEDULE_1 = frozenset({"9(2)(b)", "9(2)(g)"})


def check_register(entries: list[dict]) -> list[str]:
    """List the documentation gaps in a processing register.

    Args:
        entries: Register entries with lawful basis, Article 9 condition,
            Schedule 1 condition and policy document status.

    Returns:
        One message per problem, in register order.
```

```
    """
    problems: list[str] = []
    for entry in entries:
        name = entry["attribute"]
        if entry["lawful_basis"] not in LAWFUL_BASES:
            problems.append(f"{name}: no lawful basis")
        if entry["special"] and entry["condition"] is None:
            problems.append(f"{name}: no Article 9 condition")
        if entry["condition"] in NEEDS_SCHEDULE_1:
            if entry["schedule1"] is None:
                problems.append(f"{name}: no Schedule 1 condition")
            if not entry["apd"]:
                problems.append(f"{name}: no appropriate policy document")
    return problems
```

Encoding the register rules as a check means every new attribute is tested the same way, and the gaps show up before an auditor finds them.

E1-L3 exercise 5

Hints. (1) Count the risk factors: scale, special category data, significant effects. (2) What is a DPIA for, if not to shape the design? (3) What does Article 36 prior consultation mean?

Answer.

- Inferred health risk across 40,000 people is large-scale special category data, and ranking for promotion is profiling with significant effects. Both always need a DPIA.
- The DPIA comes before processing, so its findings can change the design, for example by dropping the health inference.
- A high risk you cannot reduce means consulting the ICO first.

Screen early, assess before processing, and consult the ICO when a high risk remains.

Common wrong answers.

- need = `not_needed`: Holding the data already changes nothing. A new use with these risk factors always needs a DPIA.
- when = `after`: A DPIA after the run cannot shape the design. It comes first.
- residual = `proceed`: An unmitigated high risk needs prior consultation with the ICO.

E2-L1 The DAMA wheel, eleven knowledge areas around governance

E2-L1 exercise 1

Hints. (1) One area decides and oversees how the others run. (2) It holds authority and control over data assets. (3) Data governance.

Answer. b) Data governance

Data governance sits at the hub and directs the ten other knowledge areas.

Common wrong answers.

- `a`: Architecture is one of the ten spokes.
- `c`: Data quality is a spoke. Governance decides the quality rules and who owns them.
- `d`: Metadata management is a spoke.

E2-L1 exercise 2

Hints. (1) Where is the problem: the report or the records? (2) Which area keeps one golden record per person? (3) Reference and master data.

Answer. b) Reference and master data Because: r1) Master data keeps one agreed record per core entity, such as a person.

Duplicate people are a master data problem. The report only exposes them.

Common wrong answers.

- `a`: The report shows the symptom. The duplicate person records cause it.
- `c`: Documents are not involved. The problem is two records for one person.
- `d`: Storage holds the rows. The rule that decides which rows are the same person belongs to master data.
- `b|r2`: Right area, wrong reason. Reference data is a list of codes. One record per person is master data.

E2-L1 exercise 3

Hints. (1) One is a list of allowed values. (2) The other describes data rather than being data. (3) `Reference`, then `Metadata`.

Answer.

```
1 Reference data: a controlled list of codes, such as grades or country
     codes
2 Metadata: data about data, such as definitions, owners and lineage
```

Reference data lists allowed codes. Metadata describes data. Master data is the agreed record itself.

Common wrong answers.

- g1 = `Master`: Master data is the agreed record of a core entity, such as a person. A list of codes is reference data.

E2-L1 exercise 4

Hints. (1) Ask what kind of failure each one is: access, meaning or completeness. (2) Where do agreed definitions live? (3) Completeness is a data quality dimension.

Answer.

- Access to salary is a data security question: masking and roles.
- A shared definition lives in the glossary, part of metadata management.
- Missing managers break a completeness rule, owned by data quality.

Naming the area first sends each problem to an owner who can fix it.

Common wrong answers.

- salary = `quality`: The salaries are correct. The problem is who can see them.
- definition = `architecture`: Layers will not settle a definition. The glossary does.
- missing = `security`: Nothing is exposed. Records are incomplete, which is a quality rule.

E2-L1 exercise 5

Hints. (1) Breadth first, then depth. (2) Who can decide what a field means? (3) Is the wheel about knowledge or about teams?

Answer.

- A full map first shows where the risk sits, not where you are comfortable.
- Named owners per area and domain turn the map into accountability. Gaps are findings.
- The wheel is a checklist. Teams can own several areas.

Map all areas, name owners, and use the wheel to route work, not to reorganise people.

Common wrong answers.

- scope = `one`: Starting where you are comfortable can leave the riskiest area unseen.

- owners = `team`: A platform team cannot own the meaning of HR data. Business owners must.
- use = `org`: The wheel organises knowledge, not people.

E2-L2 A maturity baseline you can defend

E2-L2 exercise 1

Hints. (1) Is there a documented standard? (2) Is it measured? (3) Documented and followed, but not measured.

Answer. b) 3 defined

A documented standard followed across teams is level 3, defined. Measuring it with targets would move it to 4.

Common wrong answers.

- `a`: Initial means ad hoc. A documented standard that every team follows is further along.
- `c`: Managed needs measures and targets. Nobody measures these rules yet.
- `d`: Optimised needs continuous improvement driven by measures.

E2-L2 exercise 2

Hints. (1) What would an average hide? (2) Which score stops an AI pilot in its tracks? (3) Report the minimum and name its area.

Answer. c) 0, with the area that scored it Because: r1) One weak area can sink a domain, and a mean hides it.

Report the weakest area per domain. The other scores still matter for the plan, but the minimum sets the risk.

Common wrong answers.

- `a`: The mean of 3 hides an area with no capability at all.
- `b`: The mode hides the weakest area just as the mean does.
- `d`: A baseline reports where you are today, including the gaps.
- `c|r2`: Right answer, wrong reason. The weakest area can be any of the eleven.

E2-L2 exercise 3

Hints. (1) What must exist before any score? (2) Scores before summaries. (3) Rescoring comes last, on the same scale.

Answer.

1. Agree and version the scale with stakeholders
2. Lay out the grid: knowledge areas by talent domain
3. Score each cell in whole levels, citing evidence
4. Report the weakest area per domain
5. Rescore at a fixed interval on the same scale

Scale, grid, scores with evidence, the weakest area per domain, then rescoring on the same scale.

Common wrong answers.

- Scores mean nothing until the scale is agreed. Agree it first.
- You cannot report the weakest area before you score.

E2-L2 exercise 4

Hints. (1) Validate every score before you pick the minimum. (2) Check `bool` before `int`, because `True` is an `int` in Python. (3) `min(areas.items(), key=lambda pair: (pair[1], pair[0]))` breaks ties by name.

Answer.

```python
MIN_LEVEL = 0
MAX_LEVEL = 5


def weakest_by_domain(scores: dict[str, dict[str, int]]) -> dict[str, tuple[str, int]]:
    """The weakest knowledge area and its score for each talent domain.

    Args:
        scores: Domain to knowledge area to whole-level score.

    Returns:
        Domain to (area, score). Ties go to the area first in alphabetical order.

    Raises:
        ValueError: If a score is not a whole number from 0 to 5, or a domain is empty.
    """
    result: dict[str, tuple[str, int]] = {}
    for domain, areas in scores.items():
        if not areas:
            raise ValueError(f"{domain}: no scores")
        for area, score in areas.items():
            if isinstance(score, bool) or not isinstance(score, int):
                raise ValueError(f"{domain}.{area}: score must be a whole number")
```

```
            if not MIN_LEVEL <= score <= MAX_LEVEL:
                raise ValueError(f"{domain}.{area}: score {score} is
    outside 0 to 5")
        result[domain] = min(areas.items(), key=lambda pair: (pair[1],
    pair[0]))
    return result
```

Validate first, so a typo such as 2.5 or 6 fails loudly. Then a tuple key picks the lowest score and breaks ties by name, so the result never depends on dict order.

E2-L2 exercise 5

Hints. (1) Which view shows the weakest area? (2) What can an auditor check? (3) What blocks work built on top?

Answer.

- The grid shows where the risk sits. One number cannot.
- Evidence lets anyone check a score, and rescoring use the same proof.
- Targets start with the weakest areas, because they block everything built on top.

Show the grid, cite the evidence, and set targets that fix the weakest areas first.

Common wrong answers.

- headline = `mean`: An average of 3.1 hid a 0 in this lesson's example.
- evidence = `survey`: Perception is not evidence. Cite artefacts.
- next = `pilot`: A high mean can still hide a 0 in security or ownership.

E2-L3 Governance at the hub, decision rights, policies and escalation

E2-L3 exercise 1

Hints. (1) Governance decides. Management does. (2) Which option sets a decision right? (3) Who may approve access.

Answer. b) Deciding who may approve access to salary data

Deciding who holds a decision right is governance. Running, building and granting are management.

Common wrong answers.

- `a`: Running checks is doing the work. Governance sets the thresholds and who owns them.

- `c`: Building a layer is management. Governance sets the rules it must meet.
- `d`: Granting is management. Governance decides who may approve the grant.

E2-L3 exercise 2

Hints. (1) Which one could a test check? (2) Policy says what, standard makes it measurable, procedure says how. (3) The masking rule.

Answer. b) Ratings are masked for every role except HR business partners and the line chain. Because: r1) A standard makes a policy measurable, so you can test whether it holds.

The masking rule is measurable: you can test which roles see unmasked ratings. That makes it a standard.

Common wrong answers.

- `a`: That is the policy: what must be true, and why.
- `c`: That is a procedure: how to do it.
- `d`: That is advice. Nobody can test it.
- `b|r2`: Right answer, wrong reason. A standard is defined by being measurable, not by who signs it.

E2-L3 exercise 3

Hints. (1) The top level states intent. (2) The middle level can be tested. (3) `Policy`, `Standard`, `Procedure`.

Answer.

```
1 Policy: what must be true, and why. Changes rarely, senior sign-off.
2 Standard: makes it measurable. Changes when the platform does.
3 Procedure: how to do it. Changes often.
```

Policy, standard and procedure separate intent, measure and method, so each changes at its own pace.

Common wrong answers.

- g1 = `Standard`; g2 = `Policy`: The policy states the intent. The standard makes it measurable.

E2-L3 exercise 4

Hints. (1) Who gathers the facts? (2) Who decides when owners disagree? (3) Record before you publish.

Answer.

1. The data steward gathers the definitions in use and their impact
2. The data owners of both domains try to agree
3. Without agreement, the governance council decides
4. Record the decision and its reason in the decision log
5. Update the glossary entry and tell the report owners

Steward, owners, council if needed, then the log, then the glossary and the people affected.

Common wrong answers.

- Start at the lowest level that can resolve it. The council is the last resort.
- Log the decision first, so the glossary change points to its reason.

E2-L3 exercise 5

Hints. (1) Who owns the meaning of HR data? (2) Which option can be tested? (3) Which option ends a dispute for good?

Answer.

- The business owns meaning. The platform team builds what the owner decides.
- Three levels of rules make access testable and changeable at the right pace.
- A fixed path with a log ends disputes for good.

Named owners, layered rules and a logged escalation path turn governance from advice into decisions.

Common wrong answers.

- decider = `platform`: Building a table does not give the right to define a rating. That belongs to the business owner.
- rule = `guidance`: Guidance cannot be tested or enforced.
- disputes = `meeting`: Without an owner and a log, the same dispute returns.